

Syllabus Mapping

Where Lab 1 Fits in a University Course

Agent Failure Labs University Pilot Pack

Document Purpose

This document helps instructors identify where Agent Failure Lab 1 can fit within a university cybersecurity, secure software engineering, AI security, or privacy-focused course. It shows how the lab can replace, enhance, or assess common topics in a standard semester.

1. Lab 1 Summary

Field	Description
Lab title	OpsMail Assistant: Indirect Prompt Injection via Poisoned Inbox
Primary topic	Indirect prompt injection in AI agents
Student role	Security tester or external attacker in a controlled educational cyber range
Target system	AI email assistant used for onboarding operations
Attack vector	Attacker-controlled email content
Target asset	Synthetic manager home address
Core failure	Untrusted email content is treated as instruction-like context
Student output	Trace-backed security report with root-cause analysis and mitigations
Recommended duration	60–90 minutes including briefing, lab, trace review, and short report discussion

2. What Lab 1 Teaches

Lab 1 is designed to give students hands-on experience with a modern AI agent security failure. It is not merely a prompt-writing exercise. The intended learning outcome is for students to understand how untrusted external content can influence an AI agent when that content is retrieved, placed into context, and processed alongside trusted instructions or sensitive data.

2.1 Core Concepts

- Indirect prompt injection
- AI agents and tool use
- Untrusted external content
- Trust boundaries

- Instruction/data separation
- Sensitive-data disclosure
- Least-privilege access
- Output policy enforcement
- Trace-backed security analysis
- Professional vulnerability reporting

2.2 What Students Produce

Students complete the lab and submit a short security report. A strong report includes:

- attack objective;
- threat model;
- malicious email payload;
- victim prompt or trigger action;
- trace evidence;
- observed assistant output;
- root-cause analysis;
- impact assessment;
- mitigation recommendations;
- limitations or residual risk.

3. Recommended Course Types

Course Type	Fit	Recommended Use
Introduction to Cybersecurity	Medium	Use as an emerging-threat case study or optional practical lab after students learn basic security principles.
Secure Software Engineering	High	Use during modules on secure design, input handling, least privilege, or threat modeling.
Web and Application Security	Medium	Use as a comparison to injection vulnerabilities and unsafe input handling in traditional applications.
AI Security / ML Security	Very High	Use as a primary practical lab for prompt injection, agent security, or LLM application security.

Privacy Engineering	High	Use as a synthetic PII disclosure case involving access control and output controls.
Digital Forensics / Incident Response	Medium	Use as a trace-analysis exercise where students reconstruct an AI-agent failure chain.
Cybersecurity Capstone	High	Use as a starter lab before students design their own AI security assessment or mitigation project.

4. Course Integration Modes

Agent Failure Lab 1 can be used in several ways depending on the course level and instructor goals.

Mode	Meaning
Replace	Use the lab instead of a static prompt-injection worksheet, toy demo, or purely lecture-based example.
Enhance	Use the lab after a lecture to provide applied practice and trace-backed evidence.
Assess	Use the lab as a graded practical report or take-home security write-up.
Extend	Use the lab as a bridge into broader AI security, agent security, governance, or capstone work.

5. Standard 15-Week Cybersecurity Course Mapping

The following table shows where Lab 1 can fit inside a general cybersecurity course.

Week	Typical Topic	Fit	How Lab 1 Helps
1	Security principles and threat models	Optional reference	Later lab uses confidentiality, assets, attacker capability, and trust boundaries.
2	Authentication, authorization, and access control	Enhancement	Shows why least-privilege access matters when an AI agent can access sensitive data.
3	Web security basics	Optional comparison	Allows comparison between traditional injection flaws and natural-language instruction injection.
4	Application security and input validation	Enhancement	Extends input-handling concepts to natural language and retrieved content.
5	Secure software design	Strong enhancement	Demonstrates design failure caused by weak instruction/data separation.

6	Privacy and sensitive data handling	Strong fit	Demonstrates synthetic PII disclosure and disclosure-control failures.
7	Threat modeling and abuse cases	Strong fit	Students identify attacker capability, target asset, trust boundary, and success condition.
8	AI/ML security introduction	Strong fit	Introduces AI-specific attack surfaces through a hands-on agent lab.
9	Prompt injection and LLM application security	Direct replacement or lab	Provides hands-on indirect prompt injection experience.
10	Secure logging and monitoring	Enhancement	Students use traces, tool calls, and event timelines as evidence.
11	Incident response and forensic analysis	Enhancement	Students reconstruct an exploit chain from trace evidence.
12	Risk management and governance	Enhancement	Students propose mitigations and residual-risk controls.
13	Standards, frameworks, and compliance	Enhancement	Students map the failure to AI and cybersecurity risk frameworks.
14	Student reports or presentations	Assessment artifact	Students submit a structured security report.
15	Capstone or review	Optional extension	Lab can motivate broader AI agent security projects.

6. Standard 15-Week AI Security Course Mapping

The following table shows where Lab 1 can fit inside an AI security, ML security, or trustworthy AI course.

Week	Typical Topic	Fit	How Lab 1 Helps
1	Introduction to AI security and safety	Optional preview	Provides an early example of agentic AI failure.
2	Threat modeling AI systems	Strong enhancement	Students identify assets, adversaries, trust boundaries, and success criteria.
3	Data poisoning and training-time attacks	Contrast case	Shows an inference-time/application-layer attack rather than a training-time attack.
4	Evasion and adversarial examples	Contrast case	Shows text and workflow manipulation rather than model-input perturbation.
5	Model extraction and privacy leakage	Privacy bridge	Connects AI security to sensitive-data disclosure risk.
6	LLM foundations and instruction following	Preparation	Helps students understand why instruction hierarchy matters.

7	Prompt injection and jail-breaks	Direct lab	Provides hands-on experience with malicious instruction design.
8	Indirect prompt injection and tool-using agents	Core module	Directly demonstrates attacker-controlled external content influencing an agent.
9	RAG and retrieval risks	Extension point	The same trust-boundary principles apply to retrieved documents and knowledge bases.
10	Agent tool misuse and autonomy risk	Strong enhancement	Shows how tool use increases impact when untrusted content enters context.
11	Monitoring, logging, and evaluation	Strong enhancement	Students use trace-backed evidence to explain the failure.
12	AI governance and risk management	Enhancement	Students discuss policy, disclosure controls, and residual risk.
13	Secure AI system design	Mitigation workshop	Students design controls such as least privilege and output filtering.
14	Student reports or presentations	Assessment artifact	Students submit and discuss structured security reports.
15	Capstone	Foundation	Lab can serve as the first module in a larger agent security capstone.

7. Replacement and Enhancement Guide

7.1 Lab 1 Can Replace

- a toy prompt-injection worksheet;
- a static lecture demo about LLM vulnerabilities;
- a generic “write a jailbreak prompt” exercise;
- a short written case study on AI data leakage;
- a non-interactive discussion of OWASP LLM risks.

7.2 Lab 1 Can Enhance

- a lecture on trust boundaries;
- a secure design module;
- a privacy or sensitive-data handling module;
- an AI security lecture;
- an incident-response or log-analysis session;
- a standards-mapping or risk-management discussion.

8. Suggested Pre-Lab Preparation

Students do not need deep machine learning expertise to complete Lab 1. However, the lab is most effective if students understand the following concepts before beginning.

- Basic confidentiality, integrity, and availability concepts
- Security assets and threat modeling
- Difference between trusted and untrusted input
- Basic concept of an AI agent using tools
- Basic concept of prompt injection
- Professional rules of engagement for controlled security labs

8.1 Suggested Pre-Lab Mini-Briefing

A 5–10 minute instructor briefing can cover:

1. An AI agent may read external content and use tools.
2. External content may be attacker-controlled.
3. Indirect prompt injection occurs when malicious instructions are hidden in that external content.
4. The goal is not only to exploit the system, but to explain the exploit using trace evidence.

9. Suggested Post-Lab Discussion

After students complete the lab, instructors can lead a discussion using the following prompts.

1. Why is this an indirect prompt injection rather than a direct jailbreak?
2. What was the attacker-controlled input source?
3. Where did the trust boundary fail?
4. Did the victim user directly ask for sensitive information?
5. Why did tool access increase the risk?
6. Would stricter prompt wording alone be enough to fix the issue?
7. What architectural mitigation would most directly prevent the leak?
8. How would this failure change if the agent could send emails, approve payments, or access customer records?

10. Assessment Integration

Lab 1 can be assessed in several ways.

Assessment Type	Description
Completion lab	Students complete the exploit and capture evidence.
Short report	Students submit a 2–4 page security report.
Trace analysis	Students explain the exploit chain using the event timeline and trace stream.
Mitigation memo	Students propose system-level mitigations.
Presentation	Students present the exploit path, root cause, and mitigation design.
Exam question	Instructor uses the lab as a basis for a written threat-modeling or mitigation question.

Activity	Estimated Time	Notes
Instructor preparation	20–40 min	Review handout, guide, rubric, and sample report.
Pre-lab briefing	5–10 min	Explain indirect prompt injection and rules of engagement.
Student lab time	15–25 min	Students complete hands-on lab.
Trace review discussion	10–15 min	Optional in-class review.
Report writing	30–60 min	Can be assigned as homework.
Grading	5–10 min per student	Faster if using rubric and platform traces.

11. Instructor Effort Estimate

12. Summary Recommendation

For a general cybersecurity course, Lab 1 is best placed after students have learned basic threat modeling, input handling, and confidentiality. For an AI security course, Lab 1 is best placed during the prompt injection, LLM application security, or agent security module.

The strongest adoption pattern is:

Recommended Use

Use Lab 1 as a one-session practical module that enhances or replaces a lecture-only prompt injection unit. Students complete the lab, inspect traces, and submit a short report explaining the exploit, root cause, impact, and mitigations.

13. End of Syllabus Mapping